SHEET 1 OF 5

File: sheet1of5.kicad_sch

SHEET 2 OF 5

File: sheet2of5.kicad_sch

SHEET 3 OF 5

File: sheet3of5.kicad_sch

SHEET 4 OF 5

File: sheet4of5.kicad_sch

SHEET 5 OF 5

File: sheet5of5.kicad_sch

EXTRA COMPONENTS

File: extracomponents.kicad_sch

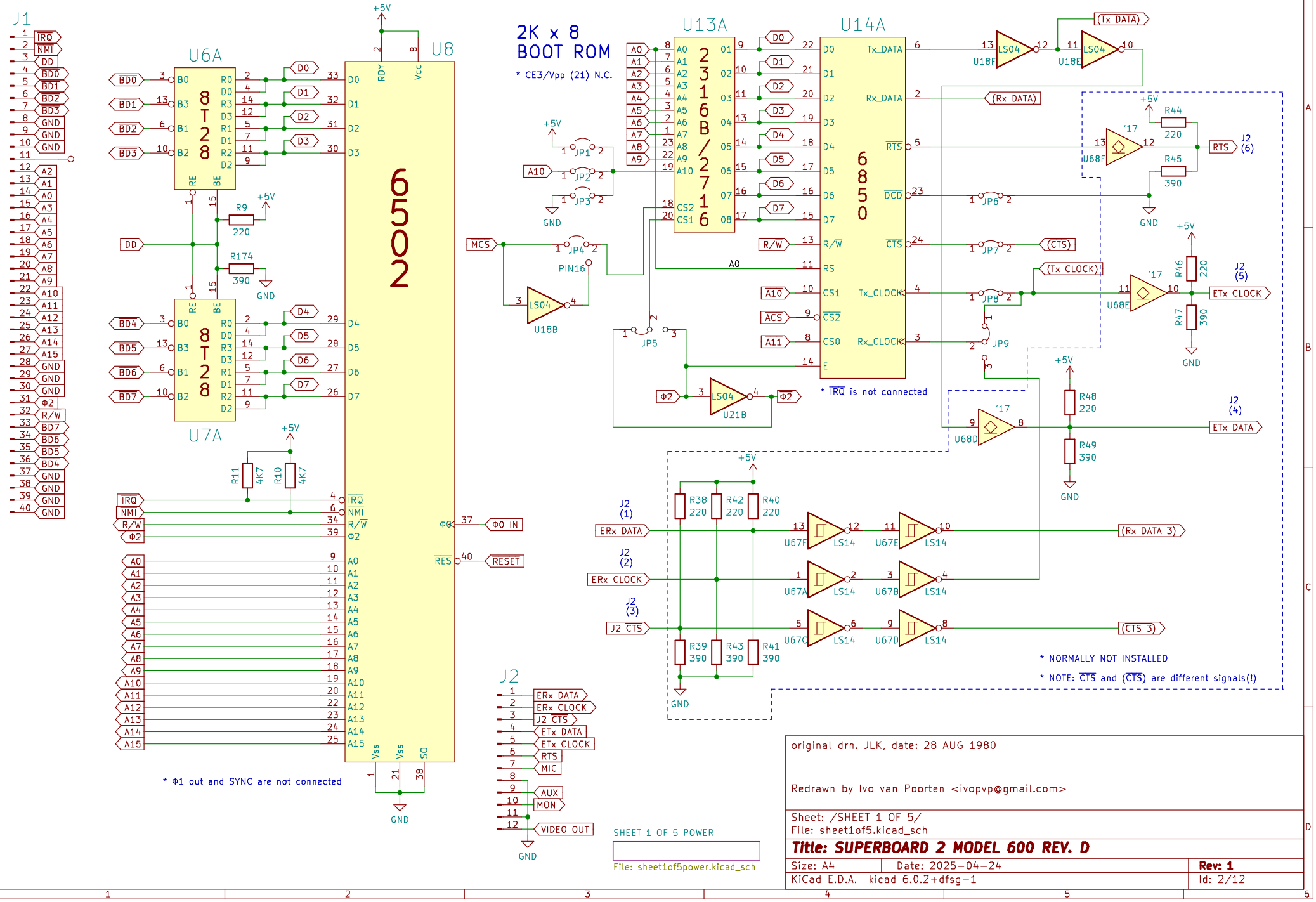
original drn. JLK, date: 28 AUG 1980

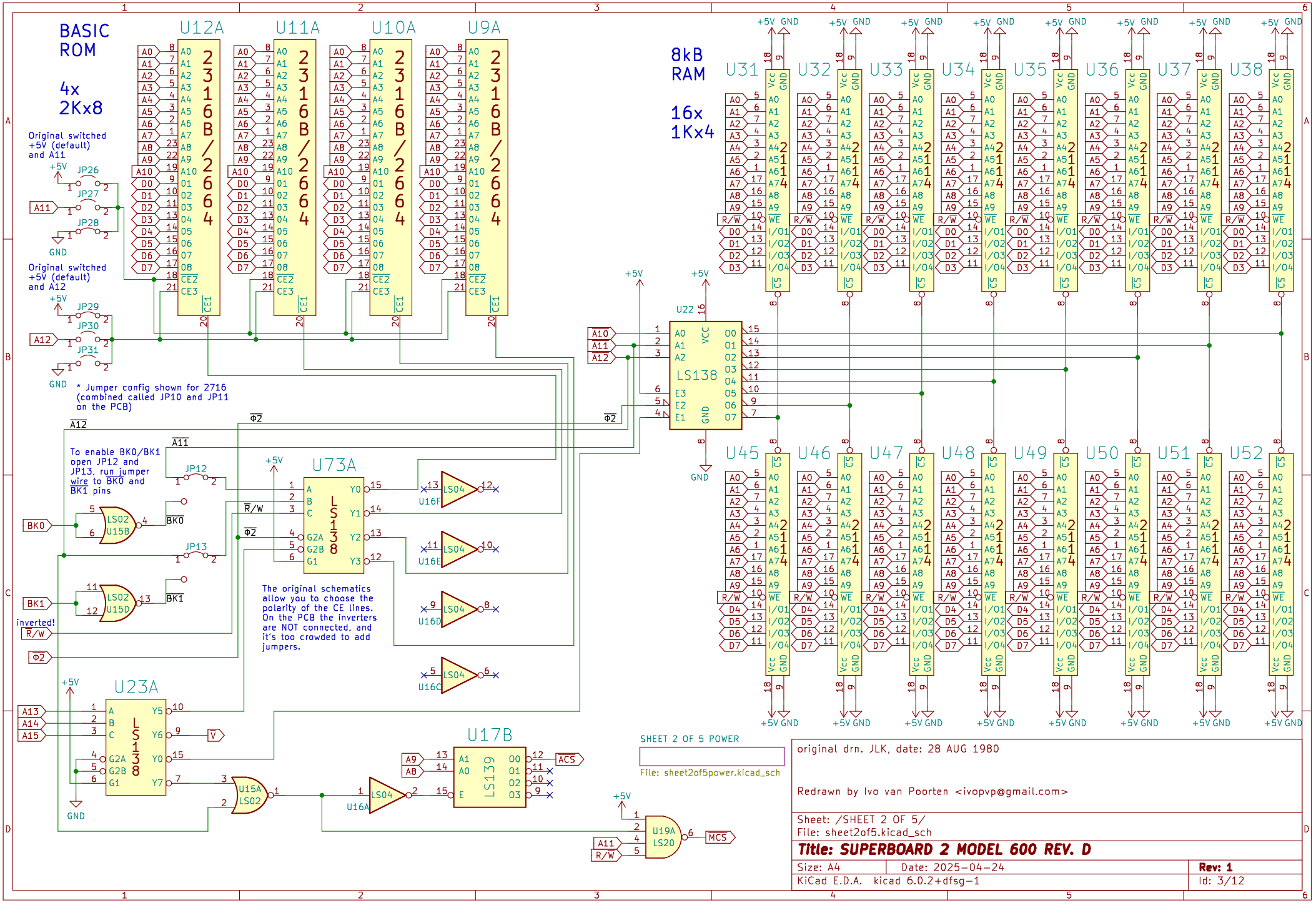
Redrawn by Ivo van Poorten <ivopvp@gmail.com>

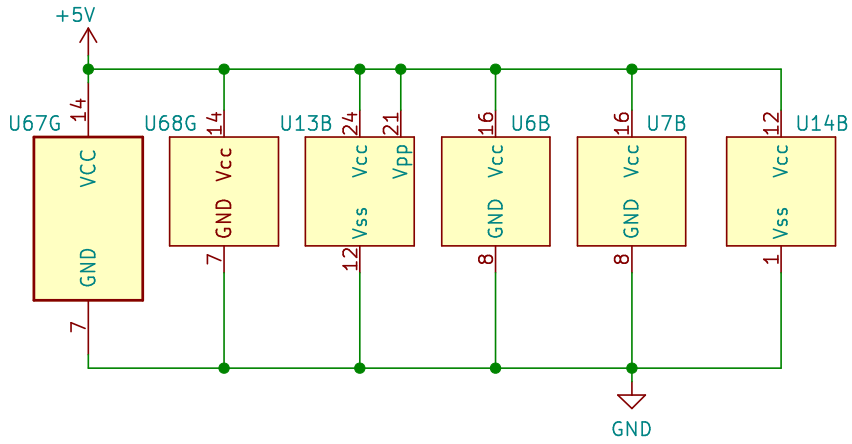
Sheet: /
File: OSI600D.kicad_sch

**Title: SUPERBOARD 2 MODEL 600 REV. D**

| Size: A5 | Date: 2025-04-24 | **Rev: 1** |
|---|---|---|
| KiCad E.D.A. kicad 6.0.2+dfsg-1 | | Id: 1/12 |


J1
U6A
U7A
U8
6502
U13A
U14A
2K x 8
BOOT ROM
* CE3/Vpp (21) N.C.
* Φ1 out and SYNC are not connected
* IRQ is not connected
* NORMALLY NOT INSTALLED
* NOTE: CTS and (CTS) are different signals(!)
J2
original drn. JLK, date: 28 AUG 1980
Redrawn by Ivo van Poorten <ivopvp@gmail.com>
Sheet: /SHEET 1 OF 5/
File: sheet1of5.kicad_sch
Title: SUPERBOARD 2 MODEL 600 REV. D
Size: A4
Date: 2025-04-24
Rev: 1
KiCad E.D.A. kicad 6.0.2+dfsg-1
Id: 2/12
SHEET 1 OF 5 POWER
File: sheet1of5power.kicad_sch


BASIC ROM
4x 2Kx8
8kB RAM
16x 1Kx4
U12A
U11A
U10A
U9A
2316B/2664
Original switched +5V (default) and A11
Original switched +5V (default) and A12
JP26
JP27
JP28
JP29
JP30
JP31
* Jumper config shown for 2716 (combined called JP10 and JP11 on the PCB)
To enable BK0/BK1 open JP12 and JP13, run jumper wire to BK0 and BK1 pins
JP12
JP13
U73A
LS138
U15B
U15D
U16F
U16E
U16D
U16C
LS04
The original schematics allow you to choose the polarity of the CE lines. On the PCB the inverters are NOT connected, and it's too crowded to add jumpers.
inverted!
U23A
U15A
U16A
U17B
LS139
U19A
LS20
U22
LS138
U31
U32
U33
U34
U35
U36
U37
U38
U45
U46
U47
U48
U49
U50
U51
U52
SHEET 2 OF 5 POWER
File: sheet2of5power.kicad_sch
original drn. JLK, date: 28 AUG 1980
Redrawn by Ivo van Poorten <ivopvp@gmail.com>
Sheet: /SHEET 2 OF 5/
File: sheet2of5.kicad_sch
Title: SUPERBOARD 2 MODEL 600 REV. D
Size: A4
Date: 2025-04-24
Rev: 1
KiCad E.D.A. kicad 6.0.2+dfsg-1
Id: 3/12



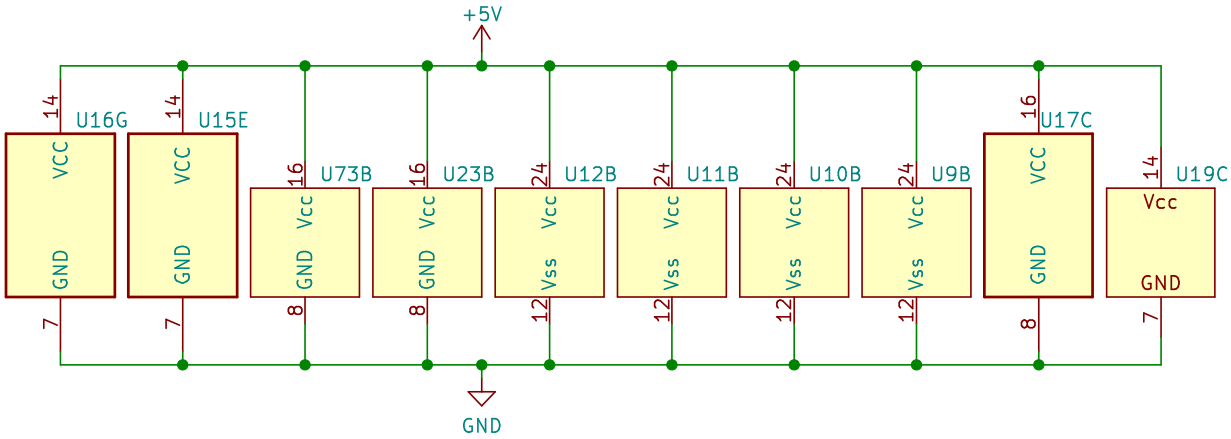

original drn. JLK, date: 28 AUG 1980

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Sheet: /SHEET 1 OF 5/SHEET 1 OF 5 POWER/
File: sheet1of5power.kicad_sch

**Title: SUPERBOARD 2 MODEL 600 REV. D**

| Size: A5 | Date: 2025-04-24 | **Rev: 1** |
|---|---|---|
| KiCad E.D.A. kicad 6.0.2+dfsg-1 | | Id: 4/12 |



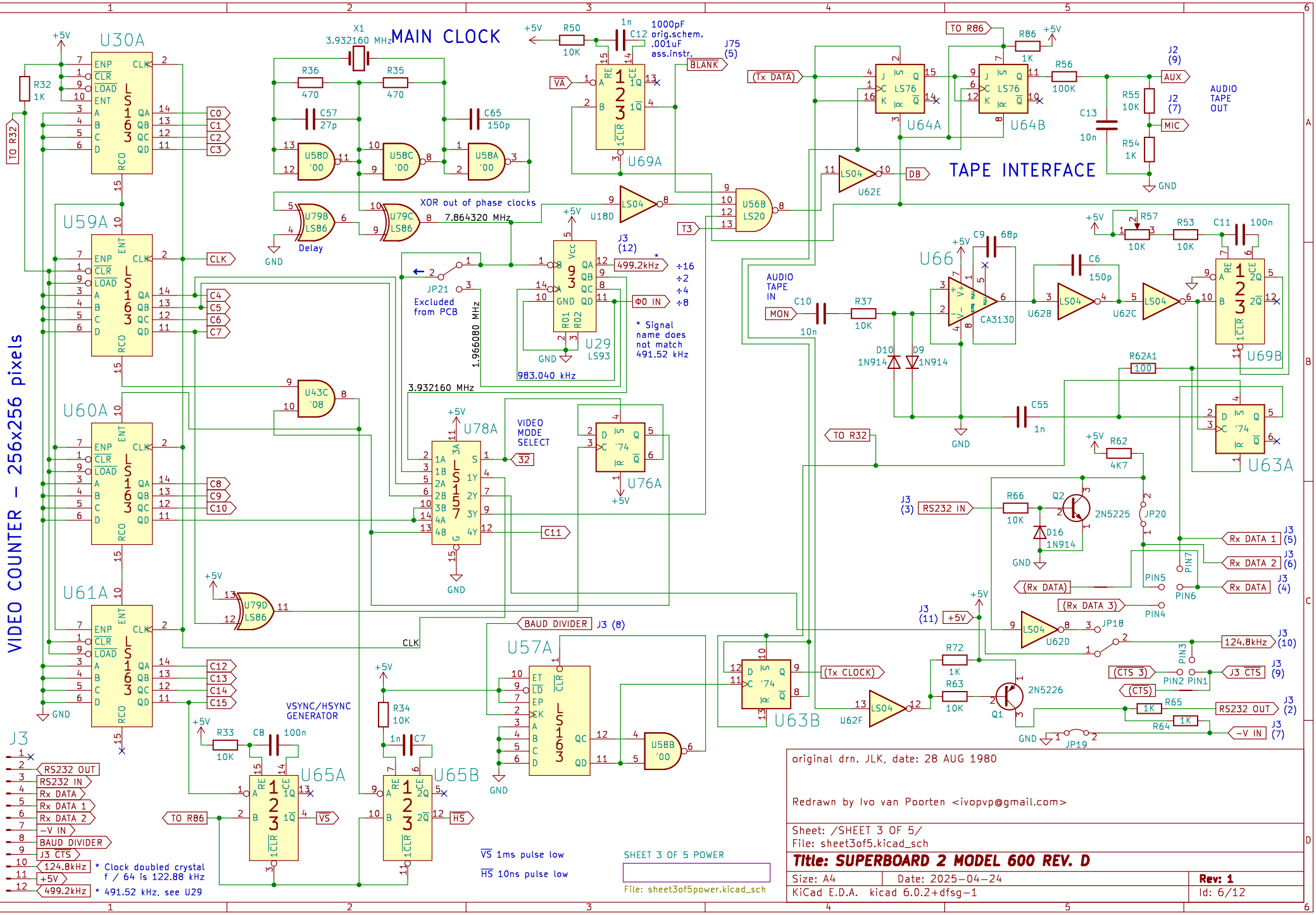

original drn. JLK, date: 28 AUG 1980

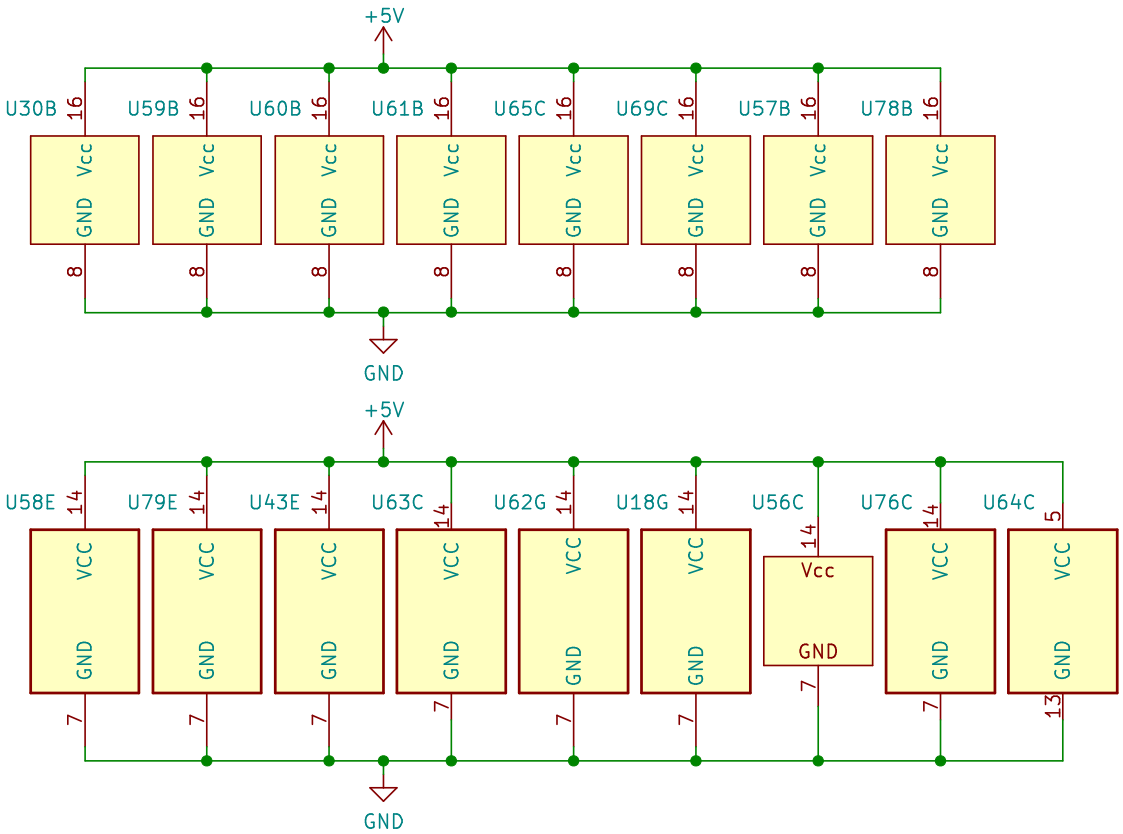
Redrawn by Ivo van Poorten <ivopvp@gmail.com>

Sheet: /SHEET 2 OF 5/SHEET 2 OF 5 POWER/
File: sheet2of5power.kicad_sch

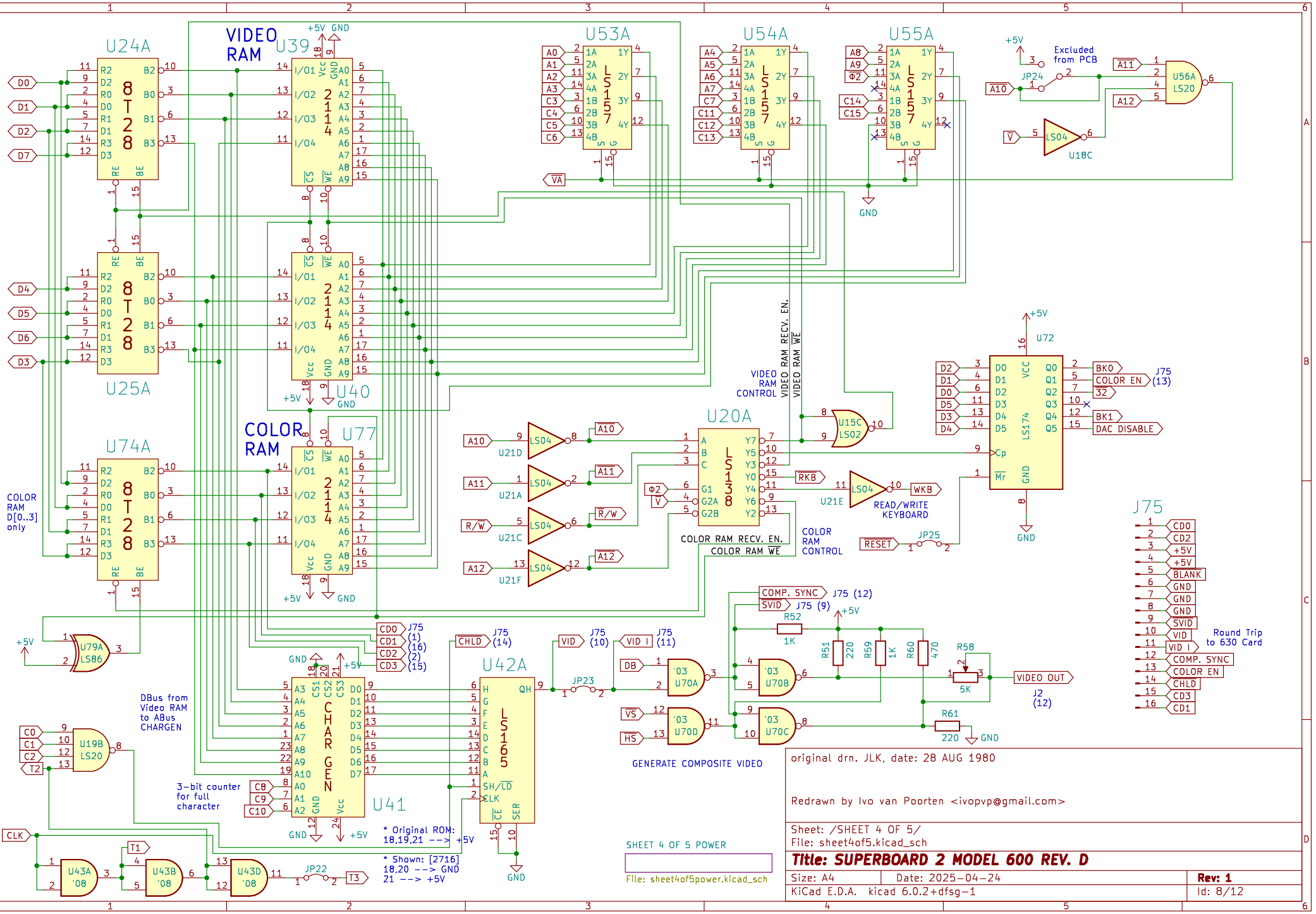
**Title: SUPERBOARD 2 MODEL 600 REV. D**

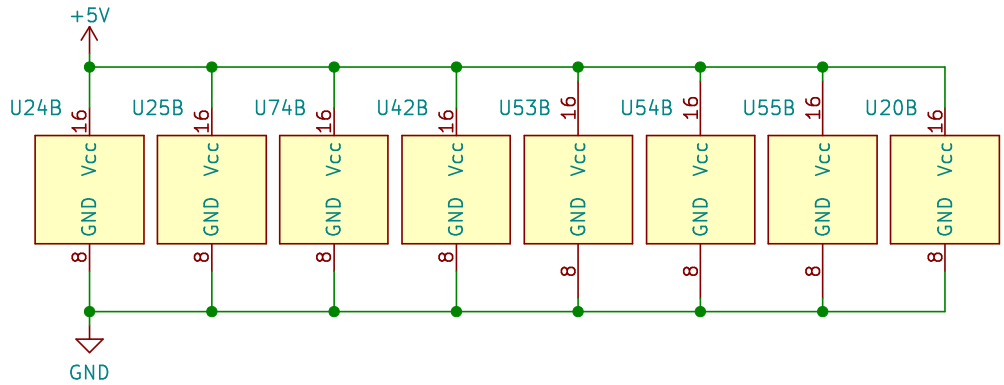
| Size: A5 | Date: 2025-04-24 | **Rev: 1** |
|---|---|---|
| KiCad E.D.A. kicad 6.0.2+dfsg-1 | | Id: 5/12 |

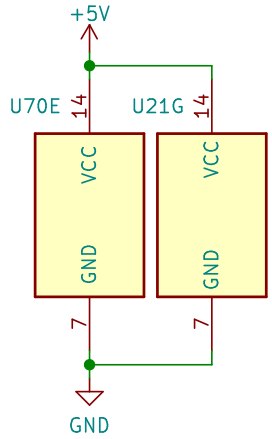

VIDEO COUNTER – 256x256 pixels
MAIN CLOCK
X1
3.932160 MHz
XOR out of phase clocks
7.864320 MHz
Delay
Excluded from PCB
1.966080 MHz
983.040 kHz
3.932160 MHz
* Signal name does not match 491.52 kHz
VIDEO MODE SELECT
VSYNC/HSYNC GENERATOR
TAPE INTERFACE
AUDIO TAPE IN
AUDIO TAPE OUT
CLK
BAUD DIVIDER
J3 (8)
VS 1ms pulse low
HS 10ns pulse low
* Clock doubled crystal f / 64 is 122.88 kHz
* 491.52 kHz, see U29
original drn. JLK, date: 28 AUG 1980
Redrawn by Ivo van Poorten <ivopvp@gmail.com>
Sheet: /SHEET 3 OF 5/
File: sheet3of5.kicad_sch
Title: SUPERBOARD 2 MODEL 600 REV. D
Size: A4
Date: 2025-04-24
Rev: 1
KiCad E.D.A. kicad 6.0.2+dfsg-1
Id: 6/12
SHEET 3 OF 5 POWER
File: sheet3of5power.kicad_sch




| | | |
|---|---|---|
| original drn. JLK, date: 28 AUG 1980 | | |
| Redrawn by Ivo van Poorten <ivopvp@gmail.com> | | |
| Sheet: /SHEET 3 OF 5/SHEET 3 OF 5 POWER/<br>File: sheet3of5power.kicad_sch | | |
| **Title: SUPERBOARD 2 MODEL 600 REV. D** | | |
| Size: A5 | Date: 2025-04-24 | **Rev: 1** |
| KiCad E.D.A. kicad 6.0.2+dfsg-1 | | Id: 7/12 |


VIDEO RAM
COLOR RAM
U24A
U25A
U74A
U39
U40
U77
U53A
U54A
U55A
U56A LS20
U18C
Excluded from PCB
JP24
U72
U20A
U15C LS02
VIDEO RAM CONTROL
VIDEO RAM RECV. EN.
VIDEO RAM WE
COLOR RAM RECV. EN.
COLOR RAM WE
COLOR RAM CONTROL
READ/WRITE KEYBOARD
RESET
JP25
J75
COLOR RAM D[0..3] only
U79A LS86
U19B LS20
U43A '08
U43B '08
U43D '08
JP22
JP23
U41
CHAR GEN
U42A
LS165
DBus from Video RAM to ABus CHARGEN
3-bit counter for full character
* Original ROM: 18,19,21 --> +5V
* Shown: [2716] 18,20 --> GND 21 --> +5V
GENERATE COMPOSITE VIDEO
COMP. SYNC
SVID
VID
VID I
CHLD
DB
VS
HS
R52
R51
R59
R60
R58
R61
VIDEO OUT
J2 (12)
Round Trip to 630 Card
SHEET 4 OF 5 POWER
File: sheet4of5power.kicad_sch
original drn. JLK, date: 28 AUG 1980
Redrawn by Ivo van Poorten <ivopvp@gmail.com>
Sheet: /SHEET 4 OF 5/
File: sheet4of5.kicad_sch
Title: SUPERBOARD 2 MODEL 600 REV. D
Size: A4
Date: 2025-04-24
Rev: 1
KiCad E.D.A. kicad 6.0.2+dfsg-1
Id: 8/12



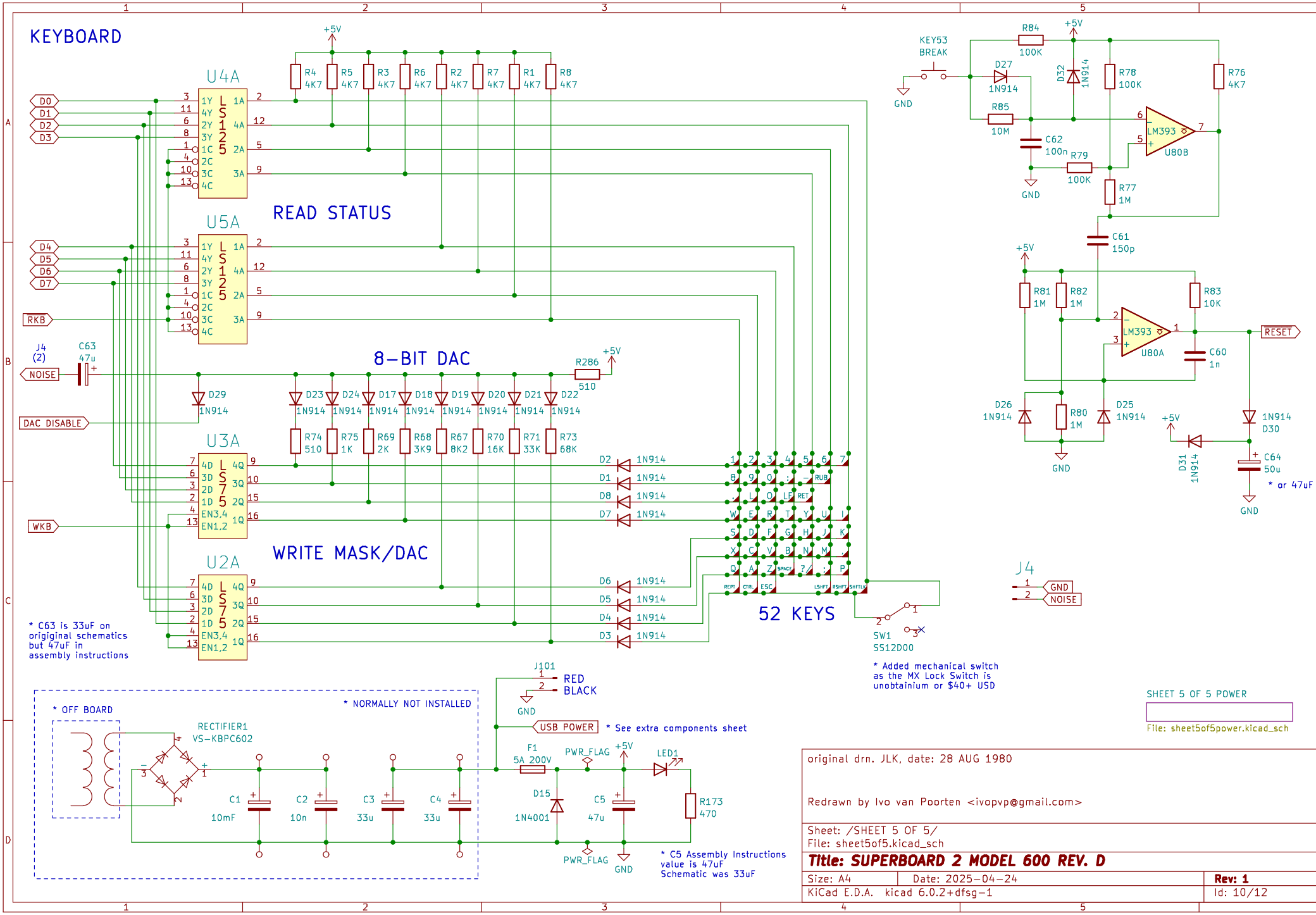

original drn. JLK, date: 28 AUG 1980

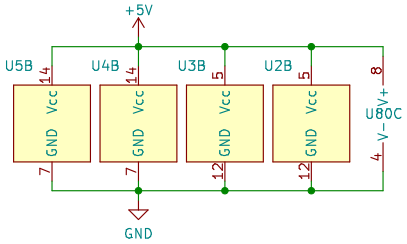
Redrawn by Ivo van Poorten <ivopvp@gmail.com>

Sheet: /SHEET 4 OF 5/SHEET 4 OF 5 POWER/
File: sheet4of5power.kicad_sch

**Title: SUPERBOARD 2 MODEL 600 REV. D**

| Size: A5 | Date: 2025-04-24 | **Rev: 1** |
|---|---|---|
| KiCad E.D.A. kicad 6.0.2+dfsg-1 | | Id: 9/12 |


KEYBOARD
U4A
U5A
U3A
U2A
74LS125
74LS75
READ STATUS
WRITE MASK/DAC
8-BIT DAC
52 KEYS
D0
D1
D2
D3
D4
D5
D6
D7
RKB
WKB
NOISE
DAC DISABLE
J4
(2)
C63
47u
R4 4K7
R5 4K7
R3 4K7
R6 4K7
R2 4K7
R7 4K7
R1 4K7
R8 4K7
+5V
R286
510
D29 1N914
D23 1N914
D24 1N914
D17 1N914
D18 1N914
D19 1N914
D20 1N914
D21 1N914
D22 1N914
R74 510
R75 1K
R69 2K
R68 3K9
R67 8K2
R70 16K
R71 33K
R73 68K
D2 1N914
D1 1N914
D8 1N914
D7 1N914
D6 1N914
D5 1N914
D4 1N914
D3 1N914
SW1
SS12D00
* Added mechanical switch as the MX Lock Switch is unobtainium or $40+ USD
* C63 is 33uF on origiginal schematics but 47uF in assembly instructions
KEY53
BREAK
GND
R84
100K
D27
1N914
D32
1N914
R78
100K
R76
4K7
R85
10M
C62
100n
R79
100K
R77
1M
LM393
U80B
C61
150p
R81
1M
R82
1M
R83
10K
LM393
U80A
C60
1n
RESET
D26
1N914
R80
1M
D25
1N914
D30
1N914
C64
50u
* or 47uF
D31
1N914
J4
GND
NOISE
* OFF BOARD
* NORMALLY NOT INSTALLED
RECTIFIER1
VS-KBPC602
C1
10mF
C2
10n
C3
33u
C4
33u
J101
RED
BLACK
USB POWER
* See extra components sheet
F1
5A 200V
PWR_FLAG
D15
1N4001
C5
47u
LED1
R173
470
* C5 Assembly Instructions value is 47uF
Schematic was 33uF
SHEET 5 OF 5 POWER
File: sheet5of5power.kicad_sch
original drn. JLK, date: 28 AUG 1980
Redrawn by Ivo van Poorten <ivopvp@gmail.com>
Sheet: /SHEET 5 OF 5/
File: sheet5of5.kicad_sch
Title: SUPERBOARD 2 MODEL 600 REV. D
Size: A4
Date: 2025-04-24
Rev: 1
KiCad E.D.A. kicad 6.0.2+dfsg-1
Id: 10/12




original drn. JLK, date: 28 AUG 1980

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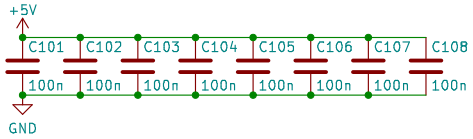
Sheet: /SHEET 5 OF 5/SHEET 5 OF 5 POWER/
File: sheet5of5power.kicad_sch

**Title: SUPERBOARD 2 MODEL 600 REV. D**

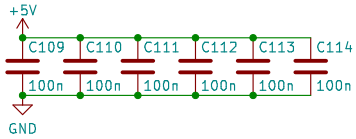
| Size: A4 | Date: 2025-04-24 | **Rev: 1** |
|---|---|---|
| KiCad E.D.A. kicad 6.0.2+dfsg-1 | | Id: 11/12 |

* Except for C8 and C11, none of the 100n bypass capacitors are shown in the original schematic. They are here, numbered C101 and up.

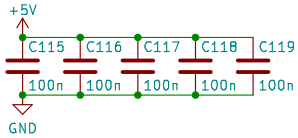
Assembly Instructions, Page 38, 8x (A)



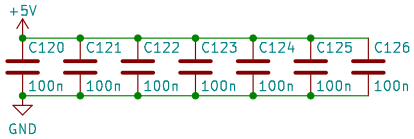

Assembly Instructions, Page 39, 6x (A)



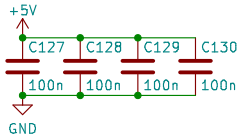

Assembly Instructions, Page 40, 5x (A)



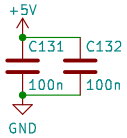

Assembly Instructions, Page 41, 7x (A)



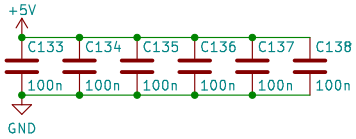

Assembly Instructions, Page 42, 4x (A)



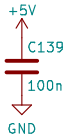

Assembly Instructions, Page 43, 2x (A)




Assembly Instructions, Page 44, 6x (A)




Assembly Instructions, Page 45, 1x (A)



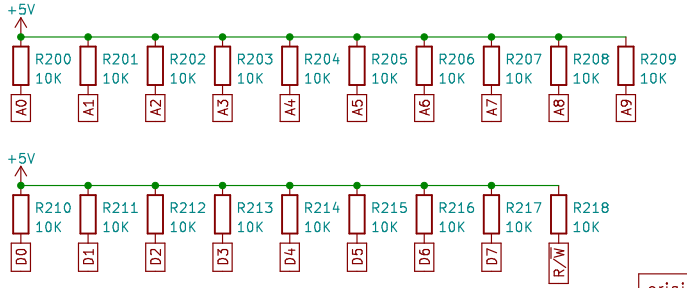

4.5mm

H1 MountingHole
H2 MountingHole
H3 MountingHole
H4 MountingHole

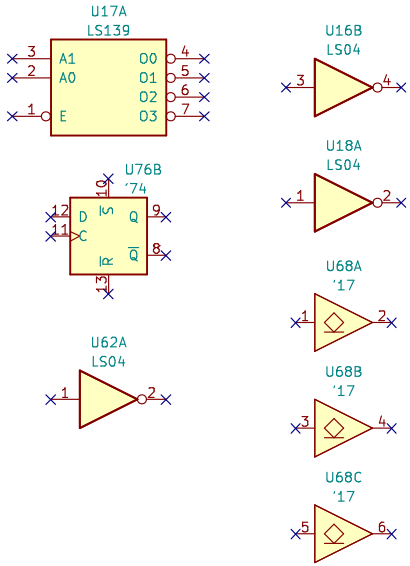
5.0mm

H5 MountingHole
H6 MountingHole
H7 MountingHole
H8 MountingHole
H9 MountingHole
H10 MountingHole
H11 MountingHole
H12 MountingHole
H13 MountingHole
H14 MountingHole
H15 MountingHole
H16 MountingHole
H17 MountingHole

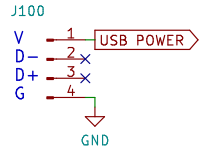
* Address and data bus pull-ups. These were not in the original schematic, but the PCB had footprints for them.




* Unconnected stuff to satisfy netlist




* Placeholder for generic USB-C 4-pin breakout board




original drn. JLK, date: 28 AUG 1980

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Sheet: /EXTRA COMPONENTS/
File: extracomponents.kicad_sch

**Title: SUPERBOARD 2 MODEL 600 REV. D**

| Size: A4 | Date: 2025-04-24 | **Rev: 1** |
|---|---|---|
| KiCad E.D.A. kicad 6.0.2+dfsg-1 | | Id: 12/12 |